

OPERATION MANUAL

SEIBOLD COMPOSER FOR AMMONIA



Pict 1: SEIBOLD COMPOSER Start Screen

Model: Ammoni Composer

SEIBOLD COMPOSER Antonio Salieri has been tested within the required Ammonia measurement range from 0.0-2.00 mg/L.

precision >97%,

error ± 5 %,

resolution 0.01 mg/L.

1. Measurement

1.1. Measurement technology

The *Measurement technology* is spectrophotometric based on Lambert-Beer's¹ law:

$$A = -\log T = -\log (I / I_0)$$

where A is absorbance, T is transmittance, I_0 is incident light, I is transmitted light. Metal is measured as a complex between the metal ions in the sample and the chelate dye. The change of the intensity of the visible light throughout cuvette containing the metal complex is directly proportional to the concentration,

$$A = \epsilon \times b \times c$$

where ϵ is the molecular absorptivity, b is the path length of the light and c is the metal concentration.

1.2. Measurement chemical requirements

a) *Buffer solution*

The role of the buffer solution is to provide a pH-controlled system and does the following:

- eliminates interference influences.
- converts oxidation stages.
- destroys complexes to free the heavy metal ions for measurement.
- dissolve suspended particles.

b) *Reagent solution (colorimetric reagent or chelate dye)*

The role of the colorimetric reagent is to form coloured complexes with the metal ions. The dye is colourless but can get to yellow/brown but this would not influence quality of measurement.

c) *Deionised water.*

d) *Standard solution of the parameter to analyse.*

1.3. Measurement steps

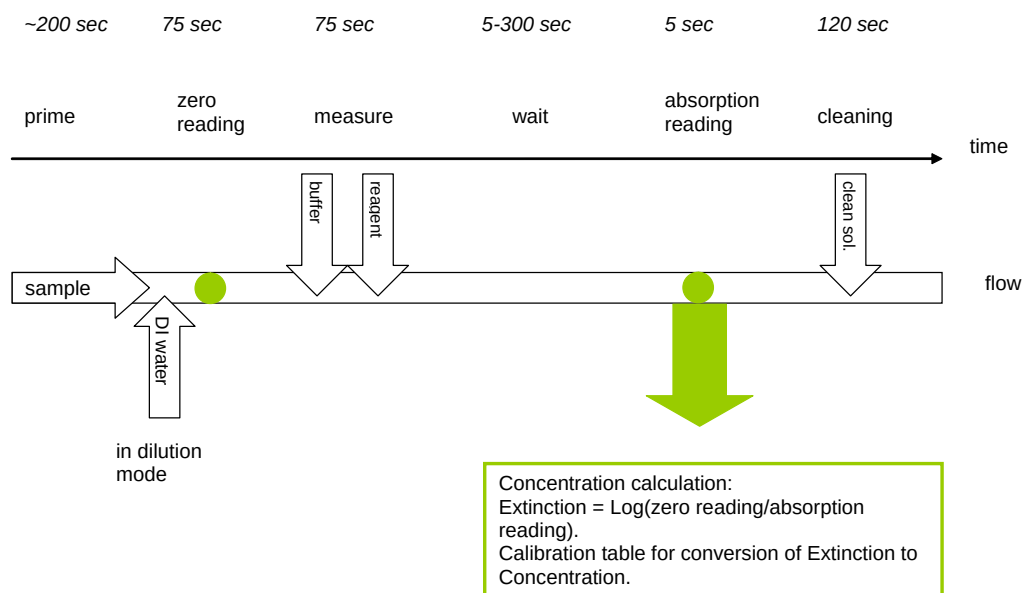
The Measurement cycle consists of 5 or more steps denoted as prime, zero, measure, wait and cleaning. In some cases additional steps for sample preparation or special cleaning are added. Each step involves the flow of a certain solution and it is defined by pump speed and measurement time. Any change could cause failure measurements.

The overall measurement time is about 10-15 minutes.

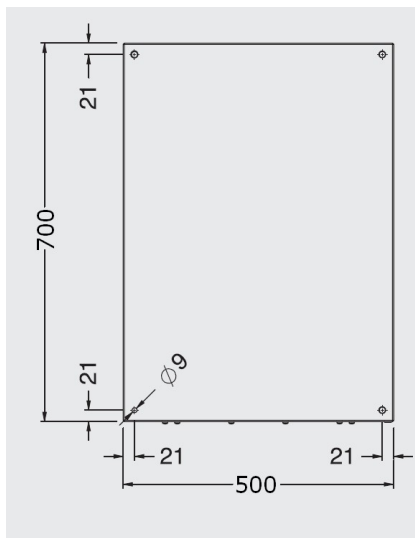
<i>Measurement step</i>	<i>Operational mode</i>	<i>Time (sec)</i>
Prime	Transfer sample to measurement unit	150-400
Zero	Collection of "zero" value	50-75
Measure	Complex formation preconditioned by buffer	50-75
Wait	Complex stabilisation	10-120
	Collection of extinction value	
Cleaning	Cleaning the system by deionised water	200

A measurement cycle lasts about 10-15 min depending on the prime value.

SEIBOLD COMPOSER MEASUREMENT DIAGRAM



2. Installation



Pict 2: Mounting Panel [mm].

The SEIBOLD Composer is a flow through system; this means all hoses are filled with water at all times. Please take care that air in the system is minimised because it could influence the measurement results.

To avoid measurement errors caused by bubbles there are three strategies embedded:

- a) *The design of measurement chamber has free space for bubbles passing by.*
- b) *High speed flushing will eliminate bubbles in the system.*
- c) *The photometer will detect bubbles through their meniscus and will not take these readings into result calculation.*

Mounting SEIBOLD COMPOSER:

The analyser cabinet is prepared for wall mounting. Please use 8mm bolts.

Below the analyser there should be space for reagents and DI water. The bottles with special caps are 40cm in height, a minimum height of 50cm is sufficient. DI water connection is recommended, clean water supply sufficient.

All hoses inlet and outlet are PTFE 6/4mm (6mm outer diameter). Please use PFA/PTFE fittings only. The fittings used in the system are SMC PFA LQ1 Series.

Pump connections:

- Sample:* Should be connected to sample line and pressure less (0.5 bar max).
Sample pump is also used to calibrate the analyser.
- Water:* The water pump is used for calibration and cleaning. DI water is recommended.
- Cleaning:* A special cleaning solution is required (10% HCL).
- Buffer:* This pump is used in the measurement step. A 2.5 L container/flask is used.
- Dye:* This pump delivers the dye to the solution. A 2.5 L container/flask is used.
- Standard:* This pump delivers the standard. A 2.5 L container/flask is used.

IMPORTANT NOTICE:

The end of the outlet (waste) hose should be about 50 cm above photometric unit and free falling (or small pressure max. 0.5 bar) to avoid sucking air into the system.

For sample inlet an overflow vessel should be used to insure pressure less inflow.

The SEIBOLD COMPOSER ensure best measurement results when measurement frequency is between 20 and 60 min. Longer intervals could need a measurement strategy adoption or adjustment.

Electronical connections:

Power: 110-240V AC, 50-60Hz.

Concentration reading: 4...20 mA line.

Connection to central system: MOD Bus on serial terminal RS485.

Relays: Alarm / Warning; Busy Fe, Busy Cl. When the analyser is connected to a central system 24V DC from central system is being used, otherwise on COMPOSER 24V (see bridge between 24(-) and COM).

Lead Analyser			
Clamps		IN	OUT
1	External Start	X	
2	Request Cleaning	X	
3	Sample I / Sample II	X	
4	Alarm		X
5	Warning		X
6	Measurement Finished		X
7	Start Sample		X
8	Busy		X
9	CH1 4...20 mA I		X
10	CH1 4...20 mA COM		X
11	RS232 RD (- RS485)	white	X
12	RS232 SD (+ RS485)	green	X
13	RS232 SG		X
14	PLC COM		X
15	PLC +		X

Pict 3: Clamps

3. Setting the SEIBOLD COMPOSER into operation

After installation, all liquids should be connected and hoses should be filled by “pumps” in “maintain” menu. Change the prime time value to the length of sample tube [calculate about 1 min per m of 6/4mm sample line].

RECOMMENDATION:

All SEIBOLD COMPOSER are delivered tested and calibrated and ready to use. Start measurements hourly for 7 days and compare regularly with laboratory measurement concentrations. If measurement results are in line with laboratory results change the sampling time to external/timer.

If results are shifted to laboratory concentrations, do a calibration. A calibration never could fix a problem as stochastic appearance of concentrations/measurement results.

Before you start:

- Check if electric connections to power and central system are established.*
- Check if outlet hoses are connected (photometer outlet must be above the photometer block free falling end).*
- Standard solutions are prepared (max concentration, min concentration, (max-min concentration)/2).*
- Test that all pumps work and filled with solutions.*

Measurement step	Sample pump	Buffer pump	Dye pump	Clean pump	Water pump
prime	✓				
zero	✓				
measure	✓	✓	✓		
wait					
clean				✓	

4. Commissioning the analyser

4.1. Calibration of the analyser

Do calibration of the analyser using a 20ppm Ammonia standard freshly prepared before the calibration (1L is sufficient for the calibration).

The calibration table consists of two columns, the 1st column displays Ammonia concentrations in ppb units and the 2nd extinction values.

Please insert the data into this Table.

Ammonia concentration (ppm)	Extinction values
0.05	
0.73	
2.67	
5.65	
12.34	
20.00	

4.2. Measurement data display

During each measurement cycle, the important measured values are stored in a table on the actual screen. The table contains zero, measure and wait values important for the calculation.

To recall these values after a series of measurements, the expert mode has to be used which is accessed by inserting the code 815, which will enable the display of all values in the menu READINGS→DATA.

4.3. Standard measurements

For the standard measurements we recommend to select two standard Ammonia values, one as Ammonia expected value of the sample and second as Ammonia alarm value. For each standard 1L of solution is enough (~ 250 ml of standard is needed per measurement). Standards have to be freshly prepared before the series of measurements.

We recommend a minimum of 3 successive measurements for data comparison.

HN3 conc (ppb)		readings	extinction	conc
Std 1	zero			
	measure			
	wait			
Std 1	zero			
	measure			
	wait			
Std 1	zero			
	measure			
	wait			
Std 2	zero			
	measure			
	wait			
Std 2	zero			
	measure			
	wait			
Std2	zero			
	measure			
	wait			

Standard measurements. For standard concentration expected Ammonia value and limit Ammonia value are selected and measured.

4.4. Sample measurements

Before running the sample measurements some important expectations have to be mentioned. In case that the sample is non-homogeneous and the distribution of its particles is not uniform, it is expected that the values of accuracy and precision will be slightly higher.

Therefore it would be preferable to work with sample that has been filtered using a corresponding pore size filter when conducting precision and accuracy tests.

It should also be taken care that the sample that is used for measurements has not been pre-treated in a chemical sense for preservation purposes. The sample should be used raw, as it was in the time of its collection.

Although the sample is stored in the dark and in the refrigerator, some chemical changes can occur over time (changes of the sample are expected even 1 day after sample collection).

Please start with filtered sample (~1L) and do 5 successive measurements.

Sample		nm 1	extinction	conc
filtered non filtered	zero			
	measure			
	wait			
filtered non filtered	zero			
	measure			
	wait			
filtered non filtered	zero			
	measure			
	wait			
filtered non filtered	zero			
	measure			
	wait			
filtered non filtered	zero			
	measure			
	wait			

4.5. Important observations during the measurements.

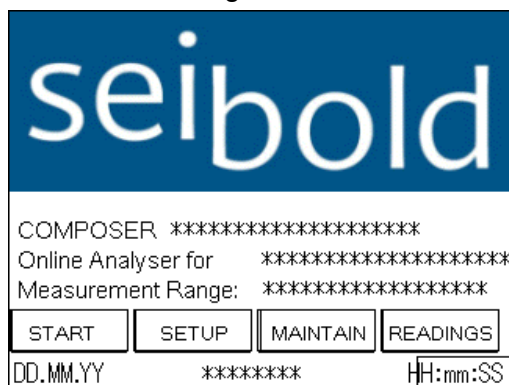
Some important observations concerning sample and functionality of the analyser have to be stated.

Sample.

- a) Turbidity
- b) Homogeneity
- c) Colour
- d) pH Value
- e) Temperature
- f) What is the colour of the mixture solution of the analyser outlet?

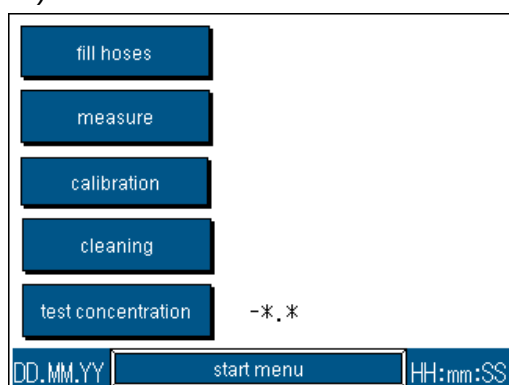
5. Operate the analyser

After switching the main button on, the first screen is the following main screen which shows the name of the analyser, measurement range and serial number.



Pict 4: Main Screen

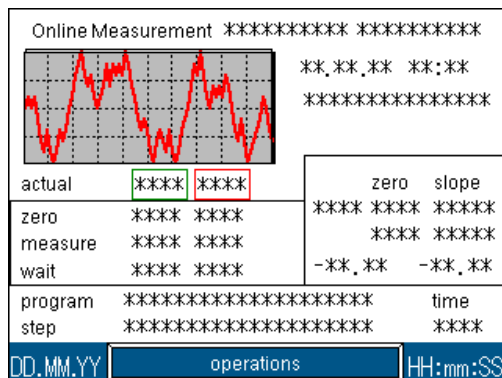
The Button "START" triggers the measurement screen with 4 options (measure, stop, calibration, test concentration).



Pict 5: Start Screen

- "measure" starts the measurement cycle.*
- "stop" ends measurement cycle.*
- "calibration" will lead to the calibration screen.*
- "test concentration" starts a measurement with defined concentration, mixed by standard solution and DI water.*

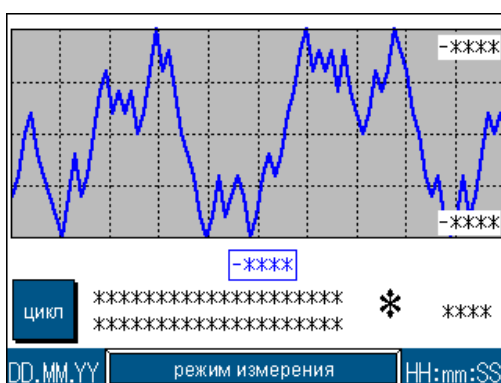
5.1. Measure



Pict 6: Operations Screen

Prime, measure and wait values are the photometer outputs taken for concentration calculation. All values in the range of 0-4000.

During the measurement, it is possible to follow measurement development in a graph mode by touching graph on the screen.



Pict 7: Measurement Development

The actual electrode and photometer value will be displayed in the coloured frame. Next to the frame is remaining seconds. Below is program name and program step. Pressing the button “cycle” will bring the operations screen up again.

6. Calibration

Calibration is needed to reference the electrode and photometer outputs to concentrations. It should be done on a regular basis (electrode: weekly; photometer; monthly) or when changes to the system are performed (changing electrode, changing electrode inner solution, changing reagents (ISA, buffer, dye), changing pump hoses).

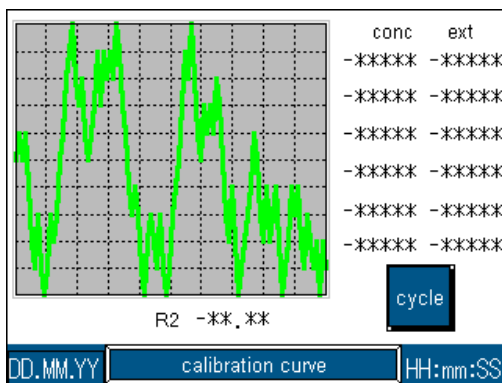
6.1. Calibration

Std. conc	-**-**	conc	ext	
standard 1	-*****	-*****	-*****	start calibration graph reset calibration show data
standard 2	-*****	-*****	-*****	
standard 3	-*****	-*****	-*****	
standard 4	-*****	-*****	-*****	
standard 5	-*****	-*****	-*****	
standard 6	-*****	-*****	-*****	

Time **** R2-**-** Reading **** Ext *****				
DD.MM.YY	calibration			HH:mm:ss

Pict 8: Calibration screen

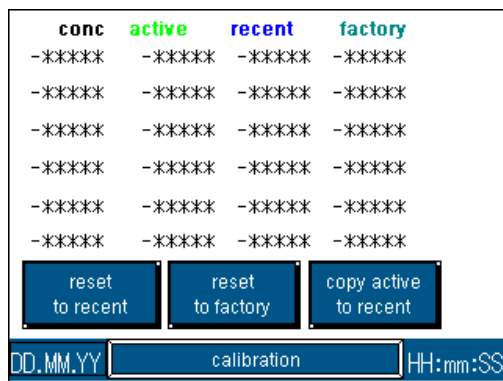
The calibration is done fully automated by a 6 point calibration. All 6 standard solutions are mixed by the analyser. The R2 (r^2) value is the *coefficient of determination* and indicates if the points are in a straight line. To see the calibration curve you can push the button "show data". Before starting the calibration, please check the standard concentration (Std. conc) is connected to the sample pump and change to the actual concentration of standard solution.



Pict 9: Calibration curve screen

"Cycle" brings you back to the operation screens.

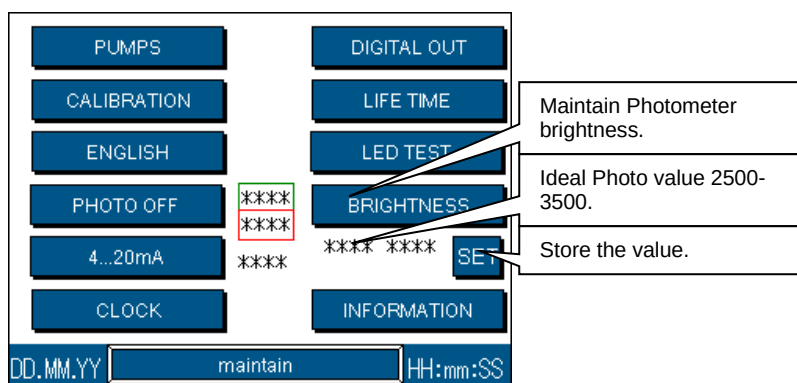
To reset calibration to factory data or recent calibration, press the button “reset calibration”.



Pict 10: Reset calibration data for Ammonia

Pressing the buttons below the data you can copy calibration tables. While active and recent may change, factory setting is fixed coded.

7. Maintain the analyser

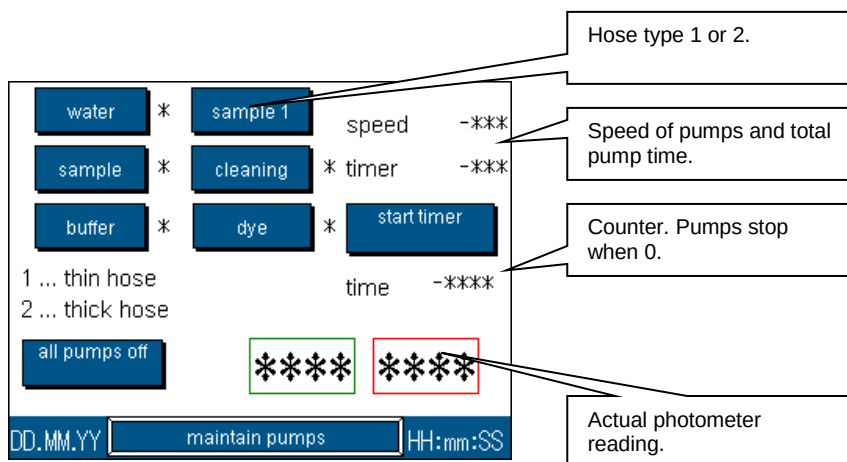


Pict 11: Maintain screen

This screen is for maintaining the analyser:

- a) Operating pumps
- b) Calibration
- c) Changing language
- d) Test the photometer and the Alarm LED
- e) Test the 4...20 mA output
- f) Digital out
- g) Check the life time
- h) Set Time and Date

7.1. Pumps



Pict 12: Pump screen

Speed indicates the pump speed for flexible pumps (Water, Standard) and timer can be used for automated turn off when time has reached timer value. Restart time by pressing "start timer" button.

Values are the direct out from photometer.

Please note that for specific purposes as standard addition the sample pump is switched to flexible mode.

<i>Pumps</i>	<i>Pump Hose</i>	<i>Jumper</i>
water	3mm	1
sample	3mm	1
cleaning	3mm	3
dye 1	1.6mm	1
dye 2	1.6mm	1
buffer	1.6mm	1

3mm hoses have a blue holder, and 1.6mm hoses are yellow.

7.2. Life Time

Pump Lifetime	in use	max use	act
Analyser in use (h)	-****.*	-****	-*
Pumps in use (h)	-****.*	-****	-*
reset in use		0 ... ignore 1 ... warning 2 ... alarm	
Consumption	in use	max use	act
Reagent (ml)	-*****	-*****	-*
reset reagent		-*.** ml/min	
DD.MM.YY	life menu		HH:mm:ss

Pict 13: Life Time screen

To minimised downtime of the analyser hose pump hoses should be changed every 6 month. This value could dramatically change to shorter periods if dilution is used (-> Water pump hose) or if sample has small particles (-> Sample pump hose). You can adjust this empiric data in life menu "max use". If no level sensors are installed the analyser can estimate the consumption of reagent and warn/alarm when reagent will be empty soon.

"Act" set up the action when limits are reached:

- 0 ... ignore
- 1 ... warning
- 2 ... alarm

7.3. Digital Out

To test the digital output signals.

Alarm	clear alarms	
Warning		
Composer busy		
Measurement ready		
Heating		
Sample 1		
**** × 1000%		
DD.MM.YY	digital out	HH:mm:ss

Pict 14: Life Time screen

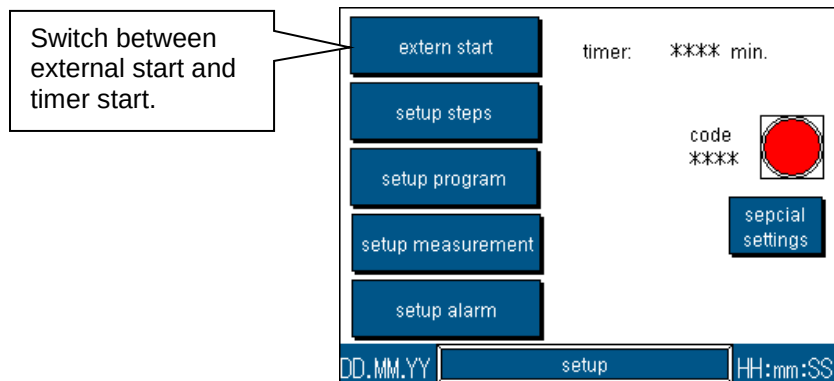
7.4. Maintenance Tasks

	Ammonia Analyser
	Change of Reagents
Every 6 Month	Clean Measurement Chamber
	Change all hose pump hoses

Duration of Tasks:

- a) *Change of Reagents:* 15 Minutes.
- b) *Clean Measurement Chamber:* 15 Minutes.
- c) *Change all hose pump hoses (6 hoses):* 30 Minutes.
- d) *“Automated tasks” are performed by analyser automatically.*

8. Setup the analyser

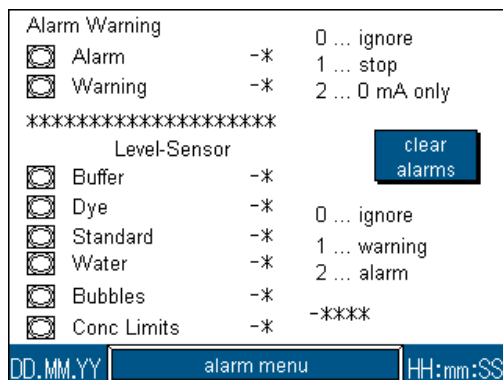


Pict 15: Setup screen

The switch “external start” controls external start (clamp 3). If in external start mode you can set the interval in minutes.

Setup steps and setup program to define measurement processes (later in this chapter more about this).

8.1. Alarm Menu



Pict 16: Setup alarm screen

Alarm and Warning signals are created by measurement range, calibration, bubbles, life time and level-sensors (if installed). Alarm and Warning signals can be ignored, cause the analyser to stop or transmit a 0 mA to the central system.

The two text lines in coloured boxes indicate the problem. Pressing “clear alarms” will clear the error state.

8.2. Setup Measurement

Concentration range.
If concentration
measured out of this
range than
warning/alarm will be
set.

low alarm	-***,***	change sample	***
high alarm	-***,***	auto dilu OFF	
standard conc	-***, **	dilu low => high	-***, *
std addition	*	dilu high => low	-***, *
		auto dilu factor	-**,*
calibration factor	-***, *	* 1 - XXXX. 2 - XXXX 3 - XXXX 4 - XXXX	
reset			
DD.MM.YY		HH:mm:SS	
setup measurement			

Pict 17: Setup measurement

Data inserted:

Alarm limits: the concentration measured must be

$\text{low alarm} < \text{concentration} < \text{high alarm}$,

otherwise an Warning/Alarm is set.

The concentration of the standard (should be the upper measurement limit).

Dilution of standard could be done automated or manually.

Calibration factor can adjust calibration to real measurements (it should be set to 1).

8.3. Programming the measurement cycle

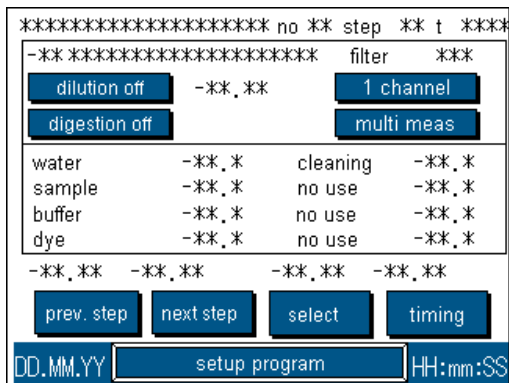
Basic Idea: A program is defined by program steps, every program step is performed for a specific time. A program step defines the pumps and pump speeds and the purpose of the step (i.e. result is taken as zero value).

***** step: **			
DU heating OFF	delay ****	char ****	cal ****
	silent ****	bubble ****	****
water	-**.*	cleaning	-**.*
sample	-**.*	no use	-**.*
buffer	-**.*	no use	-**.*
dye	-**.*	no use	-**.*
prev. step next step select			
DD.MM.YY setup steps HH:mm:ss			

Pict 18: Setup steps screen

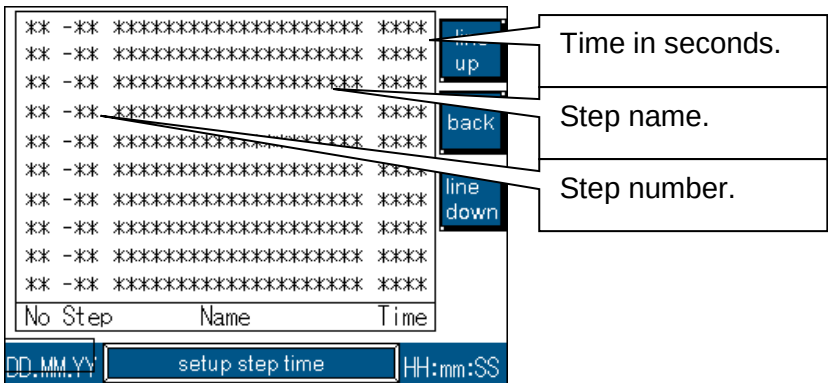
Steps can be reused in a program several times. Each and every pump speed [ml/min] is defined.

- a) *Delay: setup delay for starting reading electrode or photometer with delay.*
- b) *Silent: stops pumps before ending the step to ensure “quite” reading.*
- c) *Char: step character.*
 - a. 0 ... *not used for measurement calculation.*
 - b. 1 ... *zero value.*
 - c. 2 ... *measurement value.*
 - d. 3 ... *standard addition value.*
 - e. 4 ... *double standard addition value.*
- d) *Bubble.*
 - a. 0 ... *no bubble indication.*
 - b. 1 ... *bubble is a sample line problem.*
 - c. 2 ... *bubble is a reagent line problem.*
 - d. 3 ... *bubble is a standard solution line problem.*



Pict 19: Setup program screen

The filter value smoothens the photometer output. Typically set between 3 and 10.



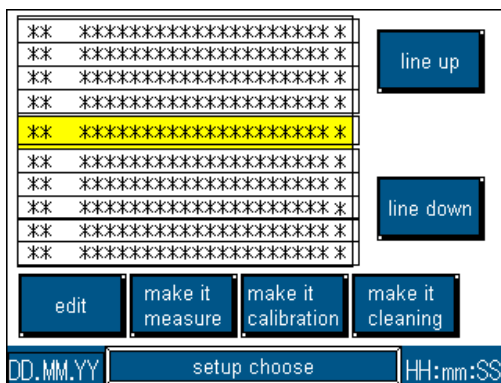
Pict 20: Setup step time screen

Please Note!

Every program may have up to 50 steps. The last step in the program must be -1 (EOP). To let the interpreter know that program is finished. A step can be used in multiple programs.

8.4. Define Standard Program

For measurement and calibration multiple programs can exist; up to 50! To mark the standard calibration and measurement program (make it standard) a special screen is used. The “choose” button will guide you there.



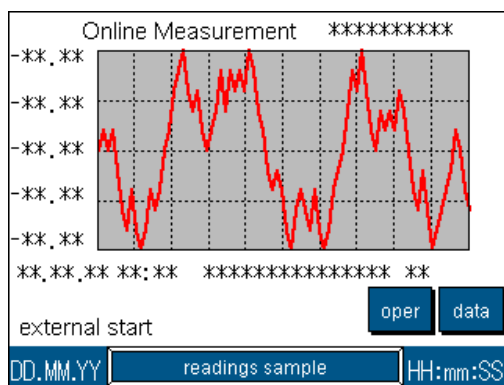
Pict 21: Define standard programs screen

Highlight the program you want to edit or make it measure/calibration or cleaning standard program. You can highlight the program by pushing “line up” and “line down” or pressing the line with program name you want to work with. Be careful when changing the program and program steps. This could lead to failures and improper measurements.

9. Measurement results

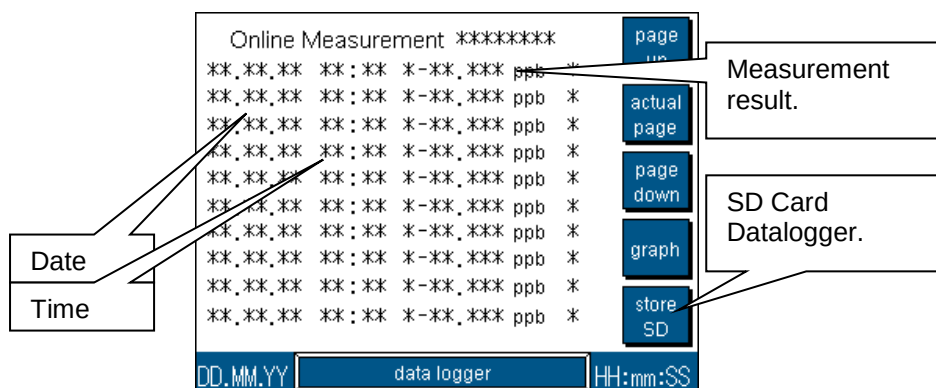
After finishing a measurement cycle the analyser does the following:

- Compare result with upper and lower limit. Out of range results in Warning/Alarm.
- Store the data in database (up to 400 measurements).
- Brings up the result screen.



Pict 22: Result screen

Pressing the “data” button opens the database and shows data logger results.



Pict 23: Datalogger screen

10. Reagents

Reagents (Buffer and Dye) for Ammonia are not toxic and not dangerous. Please note the MSDS.

The life time of reagents are 12 month in use and 60 months when stored cool (4°C) and dark.

11. Unexpected Situations, Problems?

11.1. Key data and indications.

Please observe and note key data for problem solving. Also note this data every time you are doing maintenance task.

- a) *Are hoses filled with solutions?*
- b) *Do the pumps turn?*
- c) *Photometer value On and Off; DI water and Colour complex.*
- d) *What is the colour of reagents (buffer, dye)?*
- e) *Measurements result-chain, smooth or stochastic?*
- f) *Values of measurement: zero, measurement, wait, extinction, reading.*
- g) *Standard measurement program. Measurement in dilution?*
- h) *Calibration data.*
- i) *Life data. Analyser in use, pumps in use.*
- j) *Sample condition; as usual or changes took place. Turbidity, Homogeneity, Colour, pH Value, Temperature, Bubbles, Chemistry from process, ...*
- k) *Alarm and Warning messages.*

11.2. Measurement values are not correct.

Not correct measurement values are proved by laboratory measurement or are out of measurement range. How big is the error; lower or higher? Please do step 11.1 first.

- a) *Measurement values are always at minimum limit/maximum limit.*
Compare the calibration extinction and measurement extinction. Calibration extinctions have to span more than 1000 digits between low and high concentration and have to increase their value by each concentration. Check photometer direct reading "actual" in maintain menu (PHOTOMETER ON/OFF).

- b) *Measurement values always to high or to low.*
Have sample conditions changed? Turbidity, Colour, Temperature, pH-Value and interferences by other heavy metals and chemistry (solvents, detergents ...).
- c) *Measurement values are stochastic; Big difference between two successive measurements.*
Check the sample for sample conditions as mentioned in 11.1 (bubbles, *homogeneity* ...). Check for loose electronically photometer connections and loose hose connections inside the analyser.
- d) *Measurement errors are out of stated quality (precision, accuracy).*
Check the sample conditions. Check the calibration data.

11.3. How to proceed and gather more data.

Do three measurements of sample and standard (in the same measurement range as expected sample concentration). And observe and note following key data:

- a) *Zero, Measure, Wait values.*
- b) *Bubbles in hose system?*
- c) *Colour of reaction complex between dye and sample – change of colour in measurement cycle?*
- d) *Turbidity of sample and solution mixture.*
- e) *Precision comparison between standard and sample measurements.*

All this data is needed for a good decision. In case of sample structure (bubbles, turbidity, colour, *homogeneity*) this situation has to be solved by the client.

At last a calibration could fix a wrong calibration table and fix small measurement errors.

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